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Adaptive Control of Mixed Rigid/Flexible Joint Robot Manipulators Based on Virtual Decomposition *

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Abstract

A modular approach named *virtual decomposition* is applied to adaptive control of robot manipulators with mixed rigid/flexible joints. First, the held object (payload) is decomposed from the manipulator. Then the manipulator is further decomposed into a series of rigid links and rigid/flexible joints. For each subsystem (rigid body or rigid/flexible joint), independent control design and parameter adaptation are performed, and a non-negative accompanying function is assigned. The dynamic interactions between the subsystems are completely represented by *virtual power flows* (VPFs). Both a rigid joint and a flexible joint have the same dynamic interactions with their connected links. Lyapunov asymptotic stability is guaranteed. Finally, simulation verification is conducted.

1 Introduction

Joint flexibility in robot manipulators arises from the safety requirement for accident avoidance as well as from the technical limitations on mechanical design when hydraulic drives, pneumatic drives, harmonic drives, and belts or long shafts are used [1]-[10]. When joint flexibility is considered, the control problem becomes more complicated since a fourth order dynamic model is used to replace the conventional second order dynamic model for rigid robot manipulators.

During the past ten years, a lot of contributions have been made in this area. The well-known singular perturbation technique was used for control of robot manipulators with weak joint flexibility (high joint stiffness) [1, 2, 3]. Jankowski [4] proposed a discrete inverse dynamic control strategy for flexible joint manipulators. Adaptive control of flexible joint manipulators was investigated by Marino [3], Chang [5], Lozano [6], Brogliato [7], and Yuan [8]. Lin [9] presented a unified adaptive motion/force control approach which guarantees asymptotic convergence for both motion and contact force. However, the impractical requirement on measurements of high-order derivatives limits its application.

From a practical point of view, joint flexibility for a given manipulator varies from joint to joint. Some joints exhibit considerable flexibility, while others have very weak (ignorable) flexibility and can be considered as rigid joints. This motivates research on control of mixed rigid/flexible joint robot manipulators. De Luca [10] applied nonlinear feedback linearization technique to control of mixed rigid/flexible joint manipulators with certain parameters. However, the so-called *dynamic state feedback* is still needed which makes the control algorithm quite complicated.

In this paper, a modular approach named *virtual decomposition* [11] is applied to adaptive control of a mixed rigid/flexible joint manipulator. By using *virtual decomposition*, the system is virtually decomposed into a series of rigid links, rigid joints, and flexible joints. The control problem is therefore converted into the independent control of each subsystem plus the issue of dealing with the dynamic interactions between the subsystems. It is shown that a flexible joint imposes the same dynamic interactions to its connected links as a rigid joint does. Therefore, adaptive control of each flexible joint can be carried out independently regardless of the dynamics of other subsystems. This feature greatly simplifies the control design. The cheap computation close to the recursive Newton-Euler approach enhances the capability in real-time control.

2 General Description

A. The System and Virtual Decomposition

A *six*-rigid/flexible-joint manipulator grasping a rigid object in free motion is studied in this paper, see Fig. 1. This system is virtually decomposed into the

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held object and the manipulator which is further decomposed into *six* rigid links and *six* rigid/flexible joints. Thus, only the dynamics of rigid bodies and the dynamics of rigid/flexible joints are needed in control design. *Fourteen* coordinate frames are defined as follows: frame O is fixed to the held object to describe the system motion. Frame C is located at the cutting point between the held object and the manipulator, and is fixed to both of them. Frames L_j, T_j, $j = 1, 2, \cdots, 6$, are located at the two cutting points of the jth rigid/flexible joint and are fixed to the jth link and the $(j-1)$th link, respectively, with their z axes coincident with the jth joint axis.

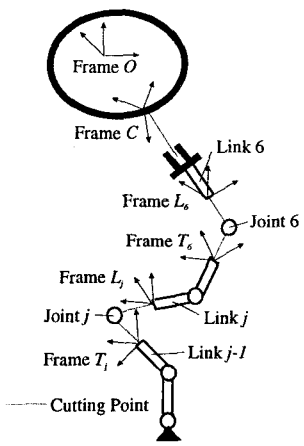


Fig. 1 The system

B. Virtual Power Flow

Definition 1: [11] The *virtual power flow* (VPF) at a given cutting point on which a moving frame α is fixed is defined as the inner product between the generalized velocity error and the generalized force error, that is

$$W_\alpha = ({}^\alpha X_r - {}^\alpha X)^T \cdot ({}^\alpha F_r - {}^\alpha F) \tag{1}$$

$(\cdot)_r$ denotes the required design vector of $(\cdot)$. ${}^\alpha X \in R^6$ denotes the generalized linear/angular velocity of the moving frame α, expressed in the frame α. ${}^\alpha F \in R^6$ denotes the generalized exerting force/moment measured and expressed in the moving frame α. i.e.

$$\begin{aligned} {}^\alpha X &= [({}^\alpha R_I \cdot \nu_\alpha)^T, ({}^\alpha R_I \cdot \omega_\alpha)^T]^T \in R^6 \\ {}^\alpha F &= [({}^\alpha R_I \cdot f_\alpha)^T, ({}^\alpha R_I \cdot m_\alpha)^T]^T \in R^6 \end{aligned}$$

where $\nu_\alpha \in R^3$ and $\omega_\alpha \in R^3$ denote the linear/angular velocities of the frame α, expressed in the inertial frame I. $f_\alpha \in R^3$ and $m_\alpha \in R^3$ denote the exerting force/moment measured in the frame α, expressed in the inertial frame I. ${}^\alpha R_I \in R^{3\times 3}$ denotes an orthogonal rotation transformation matrix which transforms a 3×1 vector expressed in the inertial frame I to that expressed in the moving frame α.

Transformations of generalized velocity and force between two moving frames α and β which are assumed to be fixed to a common rigid body are performed as

$$ {}^\beta X = {}^\alpha U_\beta^T \cdot {}^\alpha X \tag{2}$$

$$ {}^\beta F = {}^\beta U_\alpha \cdot {}^\alpha F \tag{3}$$

where ${}^\beta U_\alpha$ denotes a generalized force/moment transformation matrix which transforms a 6×1 force/moment measured and expressed in the frame α to that measured and expressed in the frame β.

C. $\mathcal{P}$ and $\underline{\mathcal{P}}$ Functions

Definition 2: $\mathcal{P}(s(t), k, a, b) \in R$ is a differentiable scalar function, where $s(t) \in R$ is a scalar variable and k, a, b are three constants with $k > 0$, $a \le b$, such that

$$\dot{\mathcal{P}} = k \cdot s(t) \cdot \kappa \tag{4}$$

$$\kappa = \begin{cases} 0 & \mathcal{P} \le a \ \& \ s(t) \le 0 \\ 0 & \mathcal{P} \ge b \ \& \ s(t) \ge 0 \\ 1 & \text{otherwise} \end{cases}$$

Lemma 1: Consider a $\mathcal{P}$ function defined in (4). For any scalar $\mathcal{P}^*$ with $a \le \mathcal{P}^* \le b$, it follows that

$$(\mathcal{P}^* - \mathcal{P})\left(s(t) - \frac{1}{k} \cdot \dot{\mathcal{P}}\right) \le 0 \tag{5}$$

Definition 3: $\underline{\mathcal{P}}(s(t), k, a, b, c) \in R$ is a second-order differentiable scalar function, where $s(t) \in R$ is a scalar variable and k, a, b, c are four constants with $k > 0$, $a \le b$, and $c > 0$, such that

$$\dot{\underline{\mathcal{P}}} = k \cdot (s(t) + c \cdot \kappa) \tag{6}$$

$$\kappa = \begin{cases} a - \underline{\mathcal{P}} & \underline{\mathcal{P}} \le a \\ b - \underline{\mathcal{P}} & \underline{\mathcal{P}} \ge b \\ 0 & \text{otherwise} \end{cases}$$

Lemma 2: Consider a $\underline{\mathcal{P}}$ function defined in (6). For any scalar $\mathcal{P}^*$ with $a \le \mathcal{P}^* \le b$, it follows that

$$(\mathcal{P}^* - \underline{\mathcal{P}})\left(s(t) - \frac{1}{k} \cdot \dot{\underline{\mathcal{P}}}\right) \le -c \cdot \kappa^2 \le 0 \tag{7}$$

$\mathcal{P}$ and $\underline{\mathcal{P}}$ functions will be used below to estimate all unknown parameters. $\mathcal{P}$ and $\underline{\mathcal{P}}$ functions have a feature that they can prevent parameter estimates from drifting, since the lower and upper bounds a and b are assigned. $[a, b]$ is an attractive real interval for any $\mathcal{P}$ and $\underline{\mathcal{P}}$ outside this interval. Therefore, the parameter adaptation by using $\mathcal{P}$ and $\underline{\mathcal{P}}$ functions is robust against unmodelling disturbances. The major difference between $\mathcal{P}$

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and $\underline{\mathcal{P}}$ functions is that $\ddot{\mathcal{P}}$ does not exist, but $\ddot{\underline{\mathcal{P}}}$ does for $\mid \dot{s}(t) \mid < +\infty$. $\mathcal{P}$ function is preferable if there is no $\ddot{\mathcal{P}}$ in the control algorithm. Otherwise, $\underline{\mathcal{P}}$ function should be chosen.

3 Adaptive Control of the Held Object

A moving reference frame D is defined to describe the desired motion of the held object. The control objective is to make the frame O tracking the frame D.

The position error between the frame D and the frame O is represented by ${}^{O}r_D \in R^3$ which denotes a vector pointing from the origin of frame O towards the origin of frame D and expressed in the frame O. The orientation error is represented by a four-parameter unit quaternion called Euler parameter $[\xi,\ {}^{O}\mu^T]^T \in R^4$ defined as

$$\begin{cases} \xi = \cos\frac{\varphi}{2} \\ {}^{O}\mu = \sin\frac{\varphi}{2} \cdot {}^{O}k \end{cases}$$

where ${}^{O}k$ denotes a unit vector $\mathbf{k}$ expressed in the frame O. The pair $(\varphi, \mathbf{k})$ is determined in such a way that rotating the frame O about $\mathbf{k}$ through φ just takes the orientation of frame D. The derivative of $[\xi,\ {}^{O}\mu^T]^T$ has been given by Wen [12] as

$$\begin{cases} \dot{\xi} &= -\frac{1}{2} \cdot {}^{O}\Delta\omega^T \cdot {}^{O}\mu \\ \frac{d}{dt}({}^{O}\mu) &= \frac{1}{2}[\xi \cdot {}^{O}\Delta\omega + ({}^{O}\Delta\omega\times) \cdot {}^{O}\mu] \end{cases} \tag{8}$$

where ${}^{O}\Delta\omega = {}^{O}R_I \cdot (\omega_D - \omega_O)$ denotes the angular velocity error expressed in the frame O.

The dynamic equation of the held object in its fixed frame O can be written as

$$M_O \cdot \frac{d}{dt}({}^{O}X) + C_O({}^{O}\omega) \cdot {}^{O}X + G_O = {}^{O}\underline{F} \tag{9}$$

where ${}^{O}\underline{F} \in R^6$ denotes the net force/moment of the held object expressed in the frame O. M_O, $C_O({}^{O}\omega)$, and G_O represent the inertial matrix, centripetal/Coriolis matrix, and gravitational vector, respectively. M_O is a constant matrix resulting from expressing velocities and forces in the object fixed frame. $C_O({}^{O}\omega)$ is a skew-symmetric matrix. The detail expression of M_O, $C_O({}^{O}\omega)$, and G_O can be found in [11].

The required net force/moment are designed by using feedforward compensation plus PD feedback as follows:

$${}^{O}\underline{F}_r = Y_{Or}^K \cdot \theta_O^K + Y_{Or} \cdot \hat{\theta}_O + K_O \cdot ({}^{O}X_r - {}^{O}X) + \begin{bmatrix} k_1 \cdot {}^{O}r_D \\ k_2 \cdot {}^{O}\mu \end{bmatrix} \tag{10}$$

where θ_O^K and θ_O denote the known and unknown parameter vectors of the held object. Y_{Or}^K and Y_{Or} are two regressor matrices defined as

$$Y_{Or}^K \cdot \theta_O^K + Y_{Or} \cdot \theta_O = M_O \cdot \frac{d}{dt}({}^{O}X_r) + C_O({}^{O}\omega) \cdot {}^{O}X_r + G_O$$

$\hat{\theta}_O$ is the estimate of θ_O. K_O is a positive-definite matrix. $k_1 > 0$ and $k_2 > 0$. ${}^{O}X = {}^{C}U_O^T \cdot J \cdot \dot{q}$. $J \in R^{6\times 6}$ and $\dot{q} \in R^6$ denote the Jacobian matrix and the joint velocity of the manipulator. ${}^{O}X_r \in R^6$, the required vector of ${}^{O}X$, is designed as

$${}^{O}X_r = diag\{{}^{O}R_I,\ {}^{O}R_I\} \cdot X_D + \lambda \cdot \begin{bmatrix} {}^{O}r_D \\ {}^{O}\mu \end{bmatrix} \tag{11}$$

where $X_D = [\nu_D^T,\ \omega_D^T]^T \in R^6$ denotes the generalized velocity of the reference frame D. $\lambda > 0$.

The unknown parameter estimate $\hat{\theta}_O$ is updated by using the $\underline{\mathcal{P}}$ function defined in (6) as

$$\left(\hat{\theta}_O\right)_\gamma = \underline{\mathcal{P}}\left((Y_{Or})_\gamma^T ({}^{O}X_r - {}^{O}X), \rho_\gamma^O, (\theta_O)_\gamma^-, (\theta_O)_\gamma^+, c_{O\gamma}\right) \tag{12}$$

where $(\hat{\theta}_O)_\gamma$ denotes the γth element of $\hat{\theta}_O$. $(Y_{Or})_\gamma$ denotes the γth column of Y_{Or}. $\rho_\gamma^O > 0$ and $c_{O\gamma} > 0$. $(\theta_O)_\gamma^-$ and $(\theta_O)_\gamma^+$ denote the lower and upper bounds of the physical parameter $(\theta_O)_\gamma$.

The non-negative accompanying function for the held object is chosen as

$$\begin{aligned} V_O &= \frac{1}{2}({}^{O}X_r - {}^{O}X)^T \cdot M_O \cdot ({}^{O}X_r - {}^{O}X) \\ &\quad + \frac{1}{2} \cdot k_1 \cdot {}^{O}r_D^T \cdot {}^{O}r_D + k_2 \cdot \left[{}^{O}\mu^T \cdot {}^{O}\mu + (1-\xi)^2\right] \\ &\quad + \frac{1}{2}(\theta_O - \hat{\theta}_O)^T \cdot \Gamma_O \cdot (\theta_O - \hat{\theta}_O) \end{aligned} \tag{13}$$

where $\Gamma_O = diag\{\frac{1}{\rho_1^O}, \cdots, \frac{1}{\rho_\gamma^O}, \cdots\}$. Differentiating (13) and applying (8)-(11), and Lemma 2 yield

$$\begin{aligned} \dot{V}_O \leq\ & -({}^{O}X_r - {}^{O}X)^T \cdot K_O \cdot ({}^{O}X_r - {}^{O}X) \\ & -k_1 \cdot \lambda \cdot {}^{O}r_D^T \cdot {}^{O}r_D - k_2 \cdot \lambda \cdot {}^{O}\mu^T \cdot {}^{O}\mu \\ & + ({}^{O}X_r - {}^{O}X)^T \cdot ({}^{O}\underline{F}_r - {}^{O}\underline{F}) \end{aligned} \tag{14}$$

Note that ${}^{O}\underline{F} = {}^{O}U_C \cdot {}^{C}F$. By designing

$${}^{C}F_r = {}^{C}U_O \cdot {}^{O}\underline{F}_r \tag{15}$$

it follows from (1) and (2) that

$$({}^{O}X_r - {}^{O}X)^T \cdot ({}^{O}\underline{F}_r - {}^{O}\underline{F}) = W_C \tag{16}$$

(16) indicates that the dynamic interaction between the held object and the manipulator is completely represented by W_C, the VPF at the end-effector of the manipulator.

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4 Adaptive Control of Rigid Links

The joint velocity of the jth joint can be written as

$$Z_j \cdot \dot{q}_j = {}^{L_j}X - {}^{T_j}U_{L_j}^T \cdot {}^{T_j}X \tag{17}$$

where $\dot{q}_j \in R$ denotes the jth joint velocity. $Z_j = [0,0,1,0,0,0]^T$ for a prismatic joint and $Z_j = [0,0,0,0,0,1]^T$ for a revolute joint.

The extended velocity vector of the *six* rigid links is expressed as

$$\mathcal{X}_m = \left[{}^{L_1}X^T, \cdots, {}^{L_j}X^T, \cdots, {}^{L_6}X^T\right]^T \in R^{36}$$

In view of (2) and (17), we have

$$\mathcal{X}_m = \mathcal{T}_m \cdot \dot{q} \tag{18}$$

$$\mathcal{X}_{mr} = \mathcal{T}_m \cdot \dot{q}_r = \mathcal{T}_m \cdot J^{-1} \cdot {}^{O}U_C^T \cdot {}^{O}X_r \tag{19}$$

where $\dot{q} = [\dot{q}_1, \cdots, \dot{q}_j, \cdots, \dot{q}_6]^T \in R^6$ denotes the joint velocities of the manipulator.

$$J = \left[\begin{array}{cccc} {}^{L_1}U_C^T \cdot Z_1 & {}^{L_2}U_C^T \cdot Z_2 & \cdots & {}^{L_6}U_C^T \cdot Z_6 \end{array}\right] \in R^{6\times 6}$$

denotes the Jacobian matrix of the manipulator, while

$$\mathcal{T}_m = \left[\begin{array}{cccc} Z_1 & 0 & \cdots & 0 \\ {}^{L_1}U_{L_2}^T \cdot Z_1 & Z_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ {}^{L_1}U_{L_6}^T \cdot Z_1 & {}^{L_2}U_{L_6}^T \cdot Z_2 & \cdots & Z_6 \end{array}\right] \in R^{36\times 6}$$

The net force/moments of the links can be written as

$${}^{L_j}\underline{F} = {}^{L_j}F - {}^{L_j}U_{T_{(j+1)}} \cdot {}^{T_{(j+1)}}F \tag{20}$$

where $j = 1,2,\cdots,6$ with ${}^{T_j}F = {}^{T_j}U_{L_j} \cdot {}^{L_j}F$ and $T_7 = C$.

By using (18), (19), and the derivative of (19), ${}^{L_j}\underline{F}_r$, $j = 1,2,\cdots,6$, the required net force/moments of the links can be obtained from (10) by replacing O with L_j and eliminating the last term in the right side. Based on (20), the required exerting force/moments are calculated in a recursive way as follows:

$$\left\{\begin{array}{rcl} {}^{L_j}F_r &=& {}^{L_j}\underline{F}_r + {}^{L_j}U_{T_{(j+1)}} \cdot {}^{T_{(j+1)}}F_r \\ {}^{T_j}F_r &=& {}^{T_j}U_{L_j} \cdot {}^{L_j}F_r \end{array}\right. \tag{21}$$

where $j = 6,5,\cdots,1$ with $T_7 = C$.

For the jth link, $j = 1,2,\cdots,6$, the corresponding non-negative accompanying function denoted as V_{L_j} and its derivative denoted as $\dot{V}_{L_j}$ can be obtained from (13) and (14), respectively, by replacing O with L_j and eliminating the second and third terms in the right sides. From (20) and (21), the last term of $\dot{V}_{L_j}$ can be written as

$$({}^{L_j}X_r - {}^{L_j}X)^T \cdot ({}^{L_j}\underline{F}_r - {}^{L_j}\underline{F}) = W_{L_j} - W_{T_{(j+1)}} \tag{22}$$

with $W_{T_7} = W_C$. It gives the dynamic interactions between the jth link and its connected joints.

5 Adaptive Control of Rigid Joints

The dynamics of the jth rigid joint including drive mechanisms can be written as

$$I_j \cdot \ddot{q}_j + f_j(q_j, \dot{q}_j) = \tau_j - Z_j^T \cdot {}^{L_j}F \tag{23}$$

where $I_j \in R$ is the equivalent rotational inertia. $f_j(q_j, \dot{q}_j) \in R$ is a term reflecting frictional force/torque. $\tau_j \in R$ is the control force/torque.

The adaptive control for the jth rigid joint is designed as

$$\tau_j = Y_{jr} \cdot \hat{\theta}_j + Z_j^T \cdot {}^{L_j}F_r + k_{vj} \cdot (\dot{q}_{jr} - \dot{q}_j) \tag{24}$$

In the right side, the first term dedicates for feedforward compensation; the second term devotes to control of the manipulator; and the third term is for velocity feedback with $k_{vj} > 0$. θ_j denotes the unknown parameter vector and Y_{jr} denotes the corresponding regressor matrix defined as

$$Y_{jr} \cdot \theta_j = I_j \cdot \ddot{q}_{jr} + f_j(q_j, \dot{q}_j)$$

The estimate of θ_j is updated by using the $\mathcal{P}$ function defined in (4) as

$$\left(\hat{\theta}_j\right)_\gamma = \mathcal{P}\left((Y_{jr})_\gamma (\dot{q}_{jr} - \dot{q}_j), \rho_{j\gamma}, (\theta_j)_\gamma^-, (\theta_j)_\gamma^+\right) \tag{25}$$

where $\rho_{j\gamma} > 0$ denotes the updating gain. $(Y_{jr})_\gamma$ and $(\theta_j)_\gamma$ denote the γth elements of Y_{jr} and θ_j, respectively. $(\theta_j)_\gamma^-$ and $(\theta_j)_\gamma^+$ denote the lower and upper bounds of the physical parameter $(\theta_j)_\gamma$.

The corresponding non-negative accompanying function is defined as

$$V_j = \frac{1}{2} \cdot I_j \cdot (\dot{q}_{jr} - \dot{q}_j)^2 + \frac{1}{2} \cdot (\theta_j - \hat{\theta}_j)^T \cdot \Gamma_j \cdot (\theta_j - \hat{\theta}_j) \tag{26}$$

where $\Gamma_j = diag\{\frac{1}{\rho_{j1}}, \cdots, \frac{1}{\rho_{j\gamma}}, \cdots\}$.

In view of (23), (24), and Lemma 1, the time derivative of (26) can be written as

$$\dot{V}_j \leq -k_{vj} \cdot (\dot{q}_{jr} - \dot{q}_j)^2 + W_{T_j} - W_{L_j} \tag{27}$$

since from (17)

$$-(\dot{q}_{jr} - \dot{q}_j) \cdot Z_j^T \cdot ({}^{L_j}F_r - {}^{L_j}F) = -W_{L_j} + W_{T_j} \tag{28}$$

(27) indicates that the dynamic interactions between the jth joint and its connected links are completely represented by $W_{T_j} - W_{L_j}$, the VPFs at the two cutting points of the joint.

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6 Adaptive Control of Flexible Joints

Without loss of generality, we assume that the jth joint is a flexible joint. The dynamic equations of the jth flexible joint can be written as

$$f_j^a(q_j, \dot{q}_j) = k_j^f \cdot (\phi_j - q_j) - Z_j^T \cdot^{L_j} F \tag{29}$$

$$I_j \cdot \ddot{\phi}_j + f_j^b(\phi_j, \dot{\phi}_j) + k_j^f \cdot (\phi_j - q_j) = \tau_j \tag{30}$$

where $k_j^f > 0$ denotes the joint elastic constant. ϕ_j denotes the equivalent rotor position. $f_j^a(q_j, \dot{q}_j)$ and $f_j^b(\phi_j, \dot{\phi}_j)$ denote the frictional terms. $f_j^a(q_j, \dot{q}_j)$ is linear in $\dot{q}_j$.

The adaptive control design is as follows

$$q_j^* = \int_0^t \dot{q}_{jr} dt + q_j(0) \tag{31}$$

$$\phi_{jd} = q_j^* + \left(Y_j^a \cdot \hat{\theta}_j^a + Z_j^T \cdot^{L_j} F_r\right) \cdot \hat{\zeta}_j^f + k_{vj}^a \cdot (\dot{q}_{jr} - \dot{q}_j) \tag{32}$$

$$\dot{\phi}_{jr} = \dot{\phi}_{jd} + \lambda_j^a \cdot (\dot{q}_{jr} - \dot{q}_j) + \lambda_j^b \cdot (\phi_{jd} - \phi_j) \tag{33}$$

$$\tau_j = D_j(t) \cdot sign(\dot{\phi}_{jr} - \dot{\phi}_j) + Y_j^b \cdot \hat{\theta}_j^b + k_{\tau j} \cdot (\dot{\phi}_{jr} - \dot{\phi}_j) + k_{pj} \cdot (\phi_{jd} - \phi_j) \tag{34}$$

where $k_{vj}^a > 0$, $\lambda_j^a > 0$, $\lambda_j^b > 0$, $k_{\tau j} > 0$, and $k_{pj} > 0$ denote feedback gains. θ_j^a, $\zeta_j^f \triangleq \frac{1}{k_j^f}$, and θ_j^b denote unknown parameters. Y_j^a and Y_j^b are two regressor matrices defined as

$$Y_j^a \cdot \theta_j^a = f_j^a(q_j, \dot{q}_j)$$

$$Y_j^b \cdot \theta_j^b = I_j \cdot \left(\lambda_j^a \cdot (\ddot{q}_{jr} - \ddot{q}_j) + \lambda_j^b \cdot (\dot{\phi}_{jd} - \dot{\phi}_j)\right) + f_j^b(\phi_j, \dot{\phi}_j) + k_j^f \cdot (\phi_j - q_j)$$

$D_j(t)$ is a positive scalar satisfying

$$D_j(t) \geq \mid I_j \cdot \ddot{\phi}_{jd} \mid \tag{35}$$

The estimates of the unknown parameters are updated by

$$\left(\hat{\theta}_j^a\right)_\gamma = \underline{\mathcal{P}}\left((Y_j^a)_\gamma(\dot{q}_{jr} - \dot{q}_j), \rho_{j\gamma}^a, (\theta_j^a)_\gamma^-, (\theta_j^a)_\gamma^+, c_{j\gamma}^a\right) \tag{36}$$

$$\hat{\zeta}_j^f = \underline{\mathcal{P}}\left((Y_j^a \cdot \hat{\theta}_j^a + Z_j^T \cdot^{L_j} F_r)(\dot{q}_{jr} - \dot{q}_j), \rho_j^f, \frac{1}{(k_j^f)^+}, \frac{1}{(k_j^f)^-}, c_j^f\right) \tag{37}$$

$$\left(\hat{\theta}_j^b\right)_\gamma = \mathcal{P}\left((Y_j^b)_\gamma(\dot{\phi}_{jr} - \dot{\phi}_j), \rho_{j\gamma}^b, (\theta_j^b)_\gamma^-, (\theta_j^b)_\gamma^+\right) \tag{38}$$

where $\rho_{j\gamma}^a > 0$, $c_{j\gamma}^a > 0$, $\rho_j^f > 0$, $c_j^f > 0$, and $\rho_{j\gamma}^b > 0$ denote updating gains. $(Y_j^a)_\gamma$, $(\theta_j^a)_\gamma$ $(Y_j^b)_\gamma$, and $(\theta_j^b)_\gamma$ denote the γth elements of Y_j^a, θ_j^a, Y_j^b, and θ_j^b, respectively. $(\cdot)^-$ and $(\cdot)^+$ denote the lower and upper bounds of the physical parameter $(\cdot)$.

In view of (10), (21), and (32), ϕ_{jd} is a linear function of $\dot{q}$. Consequently, $\dot{\phi}_{jd}$ is a linear function of $\ddot{q}$, and $\ddot{\phi}_{jd}$ is a linear function of $q^{(3)}$ and in turn a linear function of the derivative of the joint drive torques. Thus, the measurement variables in implementing (31)-(38) are q, $\dot{q}$, $\ddot{q}$, ϕ, $\dot{\phi}$, where $\ddot{q}$ can be obtained through commercially available linear accelerators mounted on the links, while D_j in (34) takes the form of

$$D_j = a_j(t)\|\psi\| + b_j(t) \tag{39}$$

so as to hold (35), where $a_j(t)$ and $b_j(t)$ are two positive scalars. $\psi = [\psi_1, \psi_2, \cdots, \psi_6]^T \in R^6$ with

$$\psi_j = \begin{cases} \dot{\phi}_j - \dot{q}_j & \text{flexible joint} \\ \dot{\tau}_j & \text{rigid joint} \end{cases}$$

where $\dot{\tau}_j$ is obtained by differentiating (24).

The corresponding non-negative accompanying function is chosen as

$$V_j = \frac{k_j^f}{k_{pj}\lambda_j^a}\left[\frac{1}{2}I_j(\dot{\phi}_{jr} - \dot{\phi}_j)^2 + \frac{1}{2}k_{pj}(\phi_{jd} - \phi_j)^2 + \frac{1}{2}\cdot(\theta_j^b - \hat{\theta}_j^b)^T \cdot \Gamma_j^b \cdot (\theta_j^b - \hat{\theta}_j^b)\right] + \frac{1}{2}\cdot k_j^f \cdot (q_j^* - q_j)^2 + \frac{1}{2}\cdot(\theta_j^a - \hat{\theta}_j^a)^T \cdot \Gamma_j^a \cdot (\theta_j^a - \hat{\theta}_j^a) + \frac{1}{2}\cdot k_j^f \cdot (\zeta_j^f - \hat{\zeta}_j^f)^2/\rho_j^f \tag{40}$$

where $\Gamma_j^a = diag\{\frac{1}{\rho_{j1}^a}, \cdots, \frac{1}{\rho_{j\gamma}^a}, \cdots\}$ and $\Gamma_j^b = diag\{\frac{1}{\rho_{j1}^b}, \cdots, \frac{1}{\rho_{j\gamma}^b}, \cdots\}$. In view of (28)-(38), Lemma 1, and Lemma 2, it follows that

$$\dot{V}_j \leq -\frac{k_{\tau j}\cdot k_j^f}{k_{pj}\cdot\lambda_j^a}\cdot(\dot{\phi}_{jr} - \dot{\phi}_j)^2 - \frac{\lambda_j^b \cdot k_j^f}{\lambda_j^a}\cdot(\phi_{jd} - \phi_j)^2 - k_{vj}^a \cdot k_j^f \cdot (\dot{q}_{jr} - \dot{q}_j)^2 + W_{T_j} - W_{L_j} \tag{41}$$

The equivalence of a flexible joint and a rigid joint in relation to the dynamic interactions to their connected links can be found by comparing (41) with (27).

7 Stability Analysis

The Lyapunov function is chosen as the sum of all non-negative accompanying functions assigned for the

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subsystems (held object, rigid links, and rigid/flexible joints).

$$V = V_O + \sum_{j=1}^{6}(V_{L_j} + V_j) \qquad (42)$$

where V_j is given in either (26) or (40). In view of (14), (16), (22), (27), (41), $W_{T_7} = W_C$, and $W_{T_1} = 0$, it gives no VPF in $\dot{V}$. Consequently, it follows that

$$\begin{aligned} \left[{}^O r_D^T, {}^O\mu^T, ({}^O X_r - {}^O X)^T\right]^T &\in L_2 \bigcap L_\infty \\ \left[(\phi_{jd} - \phi_j)^T, (\dot{\phi}_{jd} - \dot{\phi}_j)^T\right]^T &\in L_2 \bigcap L_\infty \\ & \text{for flexible joint} \end{aligned}$$

8 Simulation

The validity of the proposed approach is verified through computer simulation of a three-revolute-joint planar manipulator. The first two joints are flexible joints with joint stiffness of 1000Nm/rad, while the third joint is a rigid joint. The lengths of three links are $0.5m$, $0.5m$, and $0.2m$, respectively. The control objective is to make the manipulator tracking its desired position/orientation specified in Cartesian space as

$$\begin{aligned} X_d &= 0.8 \\ Y_d = Z_d &= -0.5 \cdot cos\left(\frac{\pi}{2}t\right) \qquad 0 \le t \le 2 \end{aligned}$$

with considerable initial errors of $[0.1(m), -0.075(m), -0.15(rad)]^T$. The position/orientation tracking errors are illustrated in Fig.2.

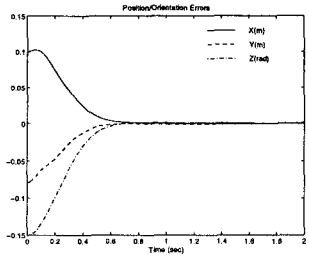


Fig. 2 Position/orientation tracking errors

9 Conclusion

Virtual decomposition approach has been applied to adaptive control of mixed rigid/flexible joint manipulators. The control problem of the complete system is converted into the control problem of each subsystem which represents the held object, each link, and each rigid/flexible joint. In the sense of Lyapunov, the dynamic interaction between every two physically connected subsystems is completely represented by the VPF between them. The VPF is independent of the dynamics of each subsystem. Therefore, a rigid joint and a flexible joint can be controlled in a unified way since they possess the same dynamic interactions to their connected links. It enhances the possibility of modular design. Each subsystem is assigned a non-negative accompanying function which particularly takes care of the dynamics of the addressed subsystem. Lyapunov function is formed just by considering the VPFs between the subsystems, without considering the dynamics of each subsystem anymore. It therefore simplifies both the control design and the stability analysis for robot manipulator with complicated dynamics. The validity of the proposed approach has been verified through computer simulation.

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